Multi-Anchor Distillation with Text-Guided Analytic Classifier for Continual Learning

Qier Meng[1], De Cheng[1*], Jiahao Li[1], and Cheng Deng[2]

[1] School of Electronic Engineering, Xidian University, Xi'an 710071, China
[2] School of Information Science and Engineering, Hohai University, Nanjing 210098

Abstract. Continual learning suffers from two forms of catastrophic forgetting: encoder drift, where visual representations shift as the model adapts to new tasks, and classifier overwriting, where gradient-based updates distort decision boundaries for previously learned classes. We present a unified exemplar-free framework that tackles both effects at the encoder and classifier levels. For the encoder, multi-anchor distillation maintains consistency with multiple historical encoder states and effectively bounds cumulative feature drift along the task sequence. For the classifier, a text-guided analytic classifier computes weights in closed form from incrementally accumulated statistics, completely bypassing gradient-based optimization. To further enhance the classifier, we incorporate a text prior that anchors classifier weights near CLIP's text embeddings to inject cross-modal semantic knowledge, and propose text subspace regularization that encourages the classifier to learn directions orthogonal to the text embedding subspace, capturing discriminative visual patterns that text semantics alone cannot represent. Experiments on CIFAR-100, ImageNet-100, ImageNet-R, and CUB-200 show state-of-the-art performance, including gains of +2.7% on CIFAR-100 and +3.5% on ImageNet-100 over prior methods. The code will be included in the supplementary material.

Keywords: Continual Learning · Class-Incremental Learning · Catastrophic Forgetting

1 Introduction

Class-incremental learning (CIL) requires models to continuously acquire knowledge of new classes while retaining the ability to recognize previously learned ones. This setting reflects real-world scenarios where data arrives sequentially and storing all previous samples is impractical due to memory or privacy constraints. The central challenge is catastrophic forgetting: neural networks tend to overwrite representations and decision boundaries of earlier classes when trained on new data, resulting in severe performance degradation on previously learned knowledge.

* Corresponding author

Traditional approaches based on regularization [7, 10], replay [16, 23], or architecture expansion [11] have made notable progress, but they operate purely in visual space without leveraging cross-modal semantic priors, which limits their ability to maintain stable class representations across tasks. The emergence of CLIP [15] has opened a promising direction: its joint image-text embedding provides stable semantic prototypes and a powerful visual backbone for efficient adaptation. Consequently, recent work has explored CLIP-based continual learning through prompt-based [18,21,22], adapter-based [4,6,28], and semantic-guided approaches [1,29]. However, existing methods typically address forgetting at a single level: prompt-based approaches [21, 22] freeze the encoder to prevent drift but limit the model's ability to learn task-specific features, while adapter-based methods [26] enable feature adaptation but the fine-tuned encoder may still drift over long task sequences. These gaps motivate a unified framework that bounds cumulative drift while ensuring classifier updates inherently free of forgetting.

Through our analysis, we identify that catastrophic forgetting in CLIP-based CIL stems from two complementary sources that must be addressed jointly. (1) Encoder drift: as the visual encoder adapts to new classes, feature representations of old classes shift, becoming entangled with new ones and misaligned with the frozen text embeddings. This drift accumulates across tasks and progressively degrades recognition of earlier classes. (2) Classifier overwriting: gradient-based optimization of the classifier modifies weights to favor new classes, corrupting decision boundaries for previously learned ones. Since the two sources compound each other, they must be addressed jointly for effective anti-forgetting.

Based on this observation, we present a unified exemplar-free framework that tackles forgetting at both the encoder and classifier levels. For the encoder, we propose *multi-anchor distillation*, which maintains consistency with multiple historical encoder states through exponentially weighted constraints. Unlike single-anchor approaches that only constrain the immediately preceding task, our method provides long-range constraints that effectively bound cumulative feature drift over extended task sequences. For the classifier, we introduce a *text-guided analytic classifier* that derives optimal weights in closed form from incrementally accumulated sufficient statistics, completely bypassing gradient-based optimization. This closed-form solution guarantees that incorporating new classes is mathematically equivalent to joint training on all data seen so far, eliminating classifier-level forgetting by construction. To further enhance the classifier, we incorporate a *text prior* that anchors classifier weights near CLIP's text embeddings, injecting cross-modal semantic knowledge to provide stable class prototypes. Additionally, we propose *text subspace regularization* that encourages the classifier to learn directions orthogonal to the text embedding subspace, enabling it to capture complementary visual patterns that text semantics alone cannot represent.

Our main contributions are summarized as follows:

- We propose multi-anchor distillation that maintains consistency with multiple historical encoder states, providing long-range constraints that effectively bound cumulative feature drift.
- We introduce a text-guided analytic classifier with text prior and text subspace regularization that bypasses gradient-based optimization entirely, substantially mitigating classifier-level forgetting while learning discriminative directions complementary to text semantics.
- Extensive experiments on four benchmarks under 10-, 20-, and 50-task settings demonstrate consistent state-of-the-art performance, with the advantage becoming more pronounced as task sequences grow longer.

2 Related Work

2.1 Anti-Forgetting Strategies in Continual Learning

Catastrophic forgetting remains the central challenge in continual learning, and existing methods can be broadly categorized into three groups based on their strategies. **Regularization methods** [7, 10] constrain parameter updates to preserve essential knowledge, either by penalizing changes to critical parameters (EWC) or applying knowledge distillation (LwF). **Replay methods** [16, 23] store or generate exemplars from previous tasks, such as iCaRL's herding-based memory buffer and DER's feature replay. **Architecture methods** [11] allocate dedicated parameters for each task to avoid interference. Following the emergence of pre-trained models, recent works leverage robust representations for continual learning. However, these methods primarily address forgetting at the encoder level through distillation or prompt isolation. We identify that forgetting also occurs at the classifier level and propose a unified framework addressing both sources.

2.2 Continual Learning with CLIP

CLIP [15] has demonstrated robust zero-shot generalization capabilities, making it an attractive backbone for continual learning. Existing CLIP-based continual learning methods can be broadly categorized into three paradigms. **Backbone fine-tuning methods** adapt CLIP's feature representations through knowledge distillation [27], model merging [12], or mixture-of-experts architectures [24]. **Prompt and adapter-based methods** freeze the backbone and introduce learnable modules. Prompt-based approaches [18, 21, 22] learn task-specific prompts, while adapter-based methods [4,6,30] append lightweight modules for cross-modal interaction or feature adaptation. **Semantic-guided methods** leverage CLIP's textual priors for knowledge transfer. SECA [1] uses textual cues to guide selective knowledge transfer and refine visual prototypes. ENGINE [29] enhances text representations using descriptions from large language models. Most existing methods focus on encoder-level forgetting while overlooking the classifier. In contrast, our work addresses anti-forgetting at both levels, with the classifier derived in closed form to completely bypass gradient-based optimization and substantially mitigate classifier-level forgetting.

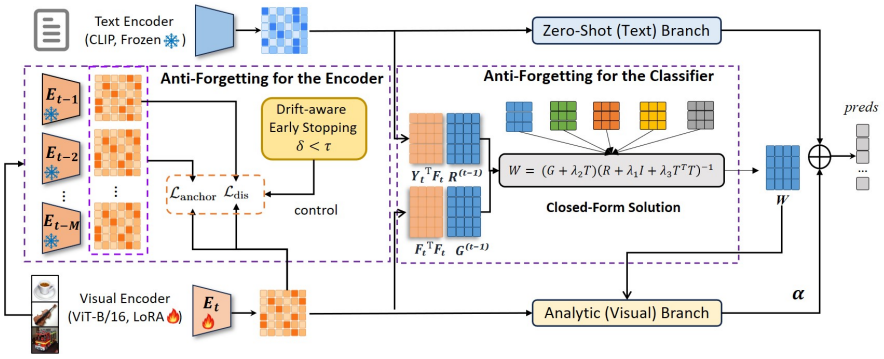


Fig. 1: Overview of our method. The framework addresses encoder drift via multi-anchor distillation with drift-aware early stopping (left), and classifier overwriting via a text-guided analytic classifier with incrementally accumulated sufficient statistics (right). Zero-shot and analytic predictions are fused at inference with mixing coefficient α (top).

2.3 Analytic Learning for Continual Learning

Analytic learning computes closed-form solutions from sufficient statistics, offering theoretical guarantees against forgetting. ACIL [34] first introduced recursive ridge regression for class-incremental learning, achieving zero forgetting by incrementally accumulating statistics under a frozen encoder. Subsequent works extended this framework: GKEAL [33] incorporates Gaussian kernel embeddings, DS-AL [32] handles data streams, GACL [31] addresses generalized CIL with mixed categories, and RanPAC [13] enhances separability via random projections. AnaCP [14] further introduces contrastive projection for feature adaptation without gradient-based training. However, existing analytic methods operate purely in visual space, ignoring the cross-modal structure of vision-language models. Our approach incorporates text embeddings as stable semantic anchors within the closed-form solution, while text subspace regularization encourages learning complementary visual directions beyond what text semantics provide.

3 Method

3.1 Overview

Our method addresses catastrophic forgetting at two levels (Figure 1). For the encoder, we adapt CLIP using LoRA and introduce *multi-anchor distillation* to maintain consistency with historical encoder states, bounding cumulative feature drift. For the classifier, we employ *analytic learning* that derives a closed-form solution from incrementally accumulated statistics, avoiding classifier-level forgetting from gradient updates. At inference, we combine the learned classifier with CLIP's text embeddings through *hybrid inference*. Sec. 3.2–3.6 detail each component.

3.2 Problem Definition and Analysis

This study addresses class-incremental learning (CIL), where tasks $\{\mathcal{T}_1, \mathcal{T}_2, \ldots, \mathcal{T}_T\}$ arrive sequentially. Each task introduces a set of new classes $\mathcal{C}_t$ with training data $\mathcal{D}_t = \{(x_i, y_i)\}_{i=1}^{N_t}$. During task t, the model has access only to $\mathcal{D}_t$; however, it must remain capable of classifying all classes encountered thus far, denoted as $\mathcal{C}_{1:t} = \bigcup_{k=1}^{t} \mathcal{C}_k$.

CLIP consists of a visual encoder f_θ and a text encoder g_ϕ that map inputs into a shared d-dimensional space. Zero-shot classification is performed by computing text prototypes $\mathbf{T} = [g_\phi(p_1), \ldots, g_\phi(p_K)]^\top$ from prompts such as "a photo of a {class}" and predicting class labels via cosine similarity. A key advantage of using CLIP for CIL is that text embeddings provide stable semantic representations for each class. These representations remain fixed throughout the learning process and do not require the storage of visual exemplars.

Catastrophic forgetting in CLIP-based continual learning arises from two primary sources. *Encoder Drift.* As the encoder adapts to new classes, the feature representations of old classes undergo drift. Previously well-separated features become entangled, and the learned representations no longer align with the frozen text embeddings. This drift accumulates across tasks and progressively degrades performance on earlier tasks. *Classifier Overwriting.* When training on new classes using gradient-based optimization, the weights of the classifier are inevitably modified. Without access to old data, these weights bias toward configurations that favor new classes, thereby corrupting the decision boundaries for previously learned classes. We address both sources through targeted anti-forgetting mechanisms: multi-anchor distillation for the encoder (Sec. 3.4) and a text-guided analytic classifier for the classification head (Sec. 3.5).

3.3 Encoder Adaptation with LoRA

Visual Encoder Adaptation via LoRA. To adapt the visual encoder while preserving the pre-trained representations, we employ Low-Rank Adaptation (LoRA) [3] on the attention projections. For each pre-trained weight matrix $\mathbf{W}_0 \in \mathbb{R}^{h \times d}$, we introduce a trainable low-rank residual:

$$\mathbf{W} = \mathbf{W}_0 + \mathbf{B}\mathbf{A}, \quad \mathbf{B} \in \mathbb{R}^{h \times r},\ \mathbf{A} \in \mathbb{R}^{r \times d}, \tag{1}$$

where $\mathbf{W}_0$ remains frozen and $r \ll \min(h, d)$. By restricting updates to this low-dimensional subspace, LoRA limits the degrees of freedom for adaptation, providing a foundation upon which our anti-forgetting mechanisms operate.

3.4 Anti-Forgetting for the Encoder

To prevent encoder drift during adaptation, we introduce three complementary mechanisms that provide constraints at different scales.

Multi-Anchor Distillation Standard feature distillation aligns the current encoder with only the immediately preceding task, which fails to account for cumulative drift: small perturbations at each task compound over time, causing features to gradually deviate from their original positions. After many tasks, this accumulated drift can significantly degrade performance on early classes, even when each individual update appears benign.

To address this limitation, we propose *multi-anchor distillation*, which maintains multiple historical states of the encoder as reference anchors throughout the learning process. Upon the completion of task t, we store the LoRA parameters as an anchor $\mathcal{A}_t$. During training on subsequent tasks, we enforce consistency with all stored anchors simultaneously:

$$\mathcal{L}_{\text{anchor}} = \sum_{i=1}^{M} w_i \, \mathbb{E}_x \Big[1 - \cos\big(f_\theta(x), f_{\mathcal{A}_i}(x) \big) \Big], \tag{2}$$

where $w_i = \gamma^{M-i}$ with $\gamma < 1$ assigns exponentially decayed importance to older anchors, and we retain a maximum of M anchors to bound memory cost.

The key insight is that multi-anchor distillation provides *long-range constraints* across the entire task sequence. By simultaneously pulling features toward multiple historical states, this mechanism prevents the encoder from drifting too far from any past configuration, thereby bounding the cumulative displacement that single-step distillation cannot control.

Feature Distillation While multi-anchor distillation bounds cumulative drift across tasks, it does not prevent abrupt changes within a single task. To ensure smooth adaptation, we additionally align the current encoder with the state from the immediately preceding task:

$$\mathcal{L}_{\text{dis}} = \mathbb{E}_x \Big[1 - \cos\big(f_\theta(x), f_{\theta^{t-1}}(x) \big) \Big]. \tag{3}$$

This provides *short-range constraints* that limit the magnitude of feature shifts during each task transition. The two distillation losses serve complementary roles: feature distillation constrains how much features can change at each step, while multi-anchor distillation constrains where features can ultimately drift to. Together, they ensure that features evolve gradually and remain well-aligned with their historical representations.

Drift-Aware Early Stopping Within a single task, the evolution of model outputs is asymmetric: early epochs primarily increase the confidence of the ground-truth class, whereas later epochs predominantly suppress the logits of negative classes, leading to excessive feature drift [5]. To prevent optimization from entering this undesirable regime, we monitor the average negative-class similarity as a drift indicator. Specifically, for each training sample x with ground-truth class y, we define the average negative-class similarity as $\overline{neg}_{\text{sim}} = \mathbb{E}_{(x,y)\sim\mathcal{D}_t} \Big[\frac{1}{|\mathcal{C}_t|-1} \sum_{k\in\mathcal{C}_t,\, k\neq y} f_\theta(x)^\top \mathbf{t}_k \Big]$, where $\mathbf{t}_k$ denotes the frozen

text embedding for class k. This quantity measures the mean cosine similarity between each training sample and all non-target text embeddings. The relative change after epoch e is:

$$\delta = \frac{\left|\overline{neg}^{e}_{\text{sim}} - \overline{neg}^{0}_{\text{sim}}\right|}{|\overline{neg}^{0}_{\text{sim}}|}. \tag{4}$$

Training for task t is terminated once δ exceeds a threshold τ. The training process proceeds in two phases: the exploration phase monitors δ to identify the optimal stopping epoch e, and the retraining phase reinitializes the LoRA parameters and trains for exactly e epochs.

Training Objective For task $t > 1$, the overall training objective combines task adaptation with anti-forgetting constraints:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_f \mathcal{L}_{\text{dis}} + \lambda_a \mathcal{L}_{\text{anchor}}, \tag{5}$$

where $\mathcal{L}_{\text{CE}}$ denotes the cross-entropy loss for the current task, and λ_f, λ_a control the strengths of the distillation terms. For the first task, we train with $\mathcal{L}_{\text{CE}}$ only, as no historical encoder states are available for distillation.

3.5 Anti-Forgetting for the Classifier: Text-guided analytic classifier

We introduce the *text-guided analytic classifier*, a closed-form classifier that simultaneously eliminates gradient-based overwriting and learns discriminative directions complementary to CLIP's text semantics. The name reflects its two defining properties: *analytic*, because classifier weights are derived in closed form from sufficient statistics rather than gradient descent; and *text-guided*, because CLIP's text embeddings serve both as a semantic prior that anchors the solution and as a reference subspace against which the classifier learns complementary visual patterns. We detail the motivation, formulation, and incremental update procedure below.

Why Analytic Learning? Gradient-based classifiers face a fundamental conflict in continual learning. Consider a linear classifier $\mathbf{W}$ trained on task t by minimizing $\mathcal{L}_t(\mathbf{W})$ via gradient descent. When task $t+1$ arrives, updates of the form $\mathbf{W} \leftarrow \mathbf{W} - \eta \nabla \mathcal{L}_{t+1}(\mathbf{W})$ inevitably modify the weights learned for previous classes, corrupting their decision boundaries. This occurs because the gradient $\nabla \mathcal{L}_{t+1}$ has no knowledge of the optimal configuration for old classes without access to their data.

Analytic learning offers a fundamentally different paradigm. Instead of iterative optimization, it computes the globally optimal solution in closed form from *sufficient statistics*, which compacts summaries that capture all the information needed to reconstruct the solution. For ridge regression, these statistics are the feature covariance $\mathbf{R} = \mathbf{F}^\top \mathbf{F}$ and class-feature correlation $\mathbf{G} = \mathbf{Y}^\top \mathbf{F}$. Crucially,

these statistics are *additive*: given data from two disjoint sets $\mathcal{D}_1$ and $\mathcal{D}_2$, we have $\mathbf{R}_{\mathcal{D}_1 \cup \mathcal{D}_2} = \mathbf{R}_{\mathcal{D}_1} + \mathbf{R}_{\mathcal{D}_2}$. This additivity enables incremental accumulation without storing raw samples. Under a stable encoder, the classifier computed after all tasks is mathematically identical to one trained jointly on the entire data seen so far, eliminating classifier-level forgetting entirely.

Why Orthogonal? While analytic learning eliminates gradient-based overwriting, the basic ridge regression solution operates purely in visual feature space, ignoring the rich cross-modal structure of CLIP. We observe that CLIP's text embeddings $\mathbf{T}$ define a semantic subspace that captures class-level concepts from language, but visual features encode richer perceptual patterns that class names cannot fully describe. A classifier confined to this text subspace would sacrifice discriminative visual cues.

To exploit both modalities, we introduce two complementary regularization terms that together define the *orthogonal* nature of our classifier:

Text Prior. We add $\lambda_2 \|\mathbf{W} - \mathbf{T}\|_F^2$ to anchor classifier weights near the frozen text embeddings $\mathbf{T}$, injecting cross-modal semantic knowledge as a stable reference point for each class.

Text Subspace Regularization. We further add

$$\mathcal{L}_{\text{subspace}} = \lambda_3 \operatorname{tr}(\mathbf{W}\mathbf{T}^\top \mathbf{T}\mathbf{W}^\top) = \lambda_3 \|\mathbf{T}\mathbf{W}^\top\|_F^2, \tag{6}$$

which penalizes the energy of each class weight $\mathbf{w}_k$ along the span of $\{\mathbf{t}_1, \ldots, \mathbf{t}_K\}$ and therefore encourages $\mathbf{W}$ to capture visual patterns that text semantics alone cannot express. The exact orthogonal projection onto the text subspace is given by $\mathbf{P} = \mathbf{T}^\top(\mathbf{T}\mathbf{T}^\top)^{-1}\mathbf{T}$, and the corresponding projection penalty is $\|\mathbf{W}\mathbf{P}\|_F^2 = \operatorname{tr}(\mathbf{W}\mathbf{T}^\top(\mathbf{T}\mathbf{T}^\top)^{-1}\mathbf{T}\mathbf{W}^\top)$. Our formulation replaces $(\mathbf{T}\mathbf{T}^\top)^{-1}$ with the identity matrix, which yields a simpler and numerically more stable surrogate. Although these two penalties weight directions inside the text subspace differently, they share the same set of minimizers and thus both steer $\mathbf{W}$ towards the orthogonal complement of the text subspace.

The two terms are complementary: the text prior provides a semantically grounded initialization, while text subspace regularization drives the classifier to learn a "residual" in the orthogonal subspace, capturing discriminative visual patterns beyond what text semantics provide. This decomposition into semantic knowledge from text and complementary cues from the orthogonal subspace is the defining characteristic of the text-guided analytic classifier and the origin of its name.

Closed-Form Solution and Incremental Update Combining all terms, the classifier objective is:

$$\mathcal{L}(\mathbf{W}) = \underbrace{\|\mathbf{Y} - \mathbf{F}\mathbf{W}^\top\|_F^2}_{\text{data fitting}} + \underbrace{\lambda_1 \|\mathbf{W}\|_F^2}_{\text{ridge}} + \underbrace{\lambda_2 \|\mathbf{W} - \mathbf{T}\|_F^2}_{\text{text prior}} + \underbrace{\lambda_3 \operatorname{tr}(\mathbf{W}\mathbf{T}^\top \mathbf{T}\mathbf{W}^\top)}_{\text{orthogonality}}, \tag{7}$$

where $\mathbf{F} \in \mathbb{R}^{N\times d}$ denotes the visual features, $\mathbf{Y} \in \mathbb{R}^{N\times K}$ is the one-hot label matrix, and $\mathbf{T} \in \mathbb{R}^{K\times d}$ contains the frozen text embeddings. Setting the derivative to zero yields:

$$\mathbf{W} = (\mathbf{G} + \lambda_2\mathbf{T})(\mathbf{R} + \lambda_1\mathbf{I} + \lambda_3\mathbf{T}^\top\mathbf{T})^{-1}, \tag{8}$$

where $\mathbf{R} = \mathbf{F}^\top\mathbf{F}$ and $\mathbf{G} = \mathbf{Y}^\top\mathbf{F}$ are sufficient statistics. The numerator combines data-driven directions ($\mathbf{G}$) with text semantics ($\lambda_2\mathbf{T}$), while the regularized inverse down-weights the text subspace through $\lambda_3\mathbf{T}^\top\mathbf{T}$, steering the solution toward orthogonal directions. This is analogous to residual learning, where the text prior provides a strong base and the orthogonal regularization captures a complementary "residual."

Since $\mathbf{W}$ depends solely on $(\mathbf{R}, \mathbf{G}, \mathbf{T})$, these statistics can be accumulated incrementally:

$$\mathbf{R}^{(t)} = \mathbf{R}^{(t-1)} + \mathbf{F}_t^\top\mathbf{F}_t, \quad \mathbf{G}^{(t)} = [\mathbf{G}^{(t-1)}; \mathbf{Y}_t^\top\mathbf{F}_t], \quad \mathbf{T}^{(t)} = [\mathbf{T}^{(t-1)}; \mathbf{T}_{\text{new}}], \tag{9}$$

requiring no storage of old samples while incorporating all historical information through the accumulated statistics.

3.6 Hybrid Inference

Because the analytic classifier $\mathbf{W}$ is trained to capture directions orthogonal to the text subspace (Sec. 3.5), its predictions encode fine-grained visual patterns but lack the coarse semantic topology that text embeddings naturally provide. We therefore re-introduce text embeddings at inference time, not to override the orthogonal signal, but to modulate it with a drift-invariant semantic prior.

Text-Guided Semantic Modulation. For each class $c \in \mathcal{C}_{1:t}$, we compute a semantic modulation coefficient:

$$k_c = \exp\Big(f_\theta(x)\,\mathbf{t}_c^\top\Big), \quad c \in \mathcal{C}_{1:t}, \tag{10}$$

where $f_\theta(x)$ is the ℓ_2-normalized visual feature, $\mathbf{t}_c$ is the frozen text embedding. Since $\mathbf{t}_c$ is fixed across tasks, k_c provides a stable semantic reference complementary to the drift-sensitive analytic classifier.

Semantically Modulated Prediction. The final prediction fuses both signals:

$$\mathbf{p} = \text{Normalize}\Big(\mathbf{k} \odot \exp\Big(\alpha \cdot \mathbf{W}\, f_\theta(x)^\top\Big)\Big), \tag{11}$$

where $\mathbf{k} = [k_1, \ldots, k_{|\mathcal{C}_{1:t}|}]$ is the semantic modulation vector, $\odot$ denotes element-wise multiplication, Normalize$(\cdot)$ rescales the input into a probability distribution, and α controls the visual logit magnitude. This design reflects a natural division of labor with Sec. 3.5: the text-guided regularization determines what $\mathbf{W}$ learns (orthogonal visual patterns), while the semantic modulation here determines how that knowledge is deployed (recombined with text semantics for drift-robust prediction).

Table 1: Comparison with state-of-the-art methods under the **10-task** setting. We report Average Accuracy (Avg) and Last Accuracy (Last) on four benchmarks. The best results are in **bold** and the second best are underlined. All methods use the ViT-B/16 backbone.

Method	Venue	CIFAR-100		ImageNet-R		ImageNet-100		CUB-200	
		Avg	Last	Avg	Last	Avg	Last	Avg	Last
Prompt-based methods									
L2P [22]	CVPR'22	82.14	73.31	81.83	76.15	80.73	67.48	77.22	68.71
DualPrompt [21]	ECCV'22	81.68	72.73	82.19	75.94	80.82	67.56	82.54	76.43
CODA-Prompt [18]	CVPR'23	77.21	62.48	78.18	67.73	64.37	35.02	84.95	80.57
SLCA [26]	ICCV'23	80.72	67.81	76.08	70.54	78.81	60.15	73.48	60.56
Aper [28]	IJCV'24	75.93	65.72	78.83	71.53	85.96	76.58	79.62	71.85
Analytic learning methods									
ACIL [34]	NeurIPS'22	80.34	72.07	79.51	72.36	81.45	72.73	75.05	66.38
RanPAC [13]	NeurIPS'23	82.62	74.83	82.08	75.62	83.34	76.05	78.52	70.47
AnaCP [14]	NeurIPS'24	84.35	77.48	83.74	78.07	85.58	78.35	81.13	73.71
CLIP-based methods									
Cont.-CLIP [19]	arXiv'22	75.33	66.87	79.30	72.18	85.14	75.58	67.21	55.03
PROOF [30]	TPAMI'25	85.05	76.47	83.02	77.22	84.89	72.67	83.28	75.72
CLAP [6]	NeurIPS'24	86.30	78.42	85.93	80.16	87.93	79.35	83.72	76.10
RAPF [4]	ECCV'24	86.38	79.22	85.76	80.46	87.68	80.42	83.22	76.53
MOE4CL [24]	CVPR'24	85.54	78.56	85.46	80.94	86.57	76.84	83.89	76.18
MagMax [12]	ECCV'24	85.80	79.18	87.31	81.03	86.51	76.10	84.15	75.95
ENGINE [29]	ICCV'25	87.10	79.41	86.40	80.55	86.62	78.83	84.42	76.28
MG-CLIP [5]	ICCV'25	87.18	80.74	87.75	82.85	87.49	78.56	84.73	76.55
BOFA [9]	AAAI'26	86.68	79.52	85.60	79.80	84.67	81.41	85.38	76.40
CLG-CBM [25]	CVPR'25	84.67	77.04	83.39	77.40	87.01	79.15	85.58	**82.38**
SECA [1]	AAAI'26	87.97	81.76	88.92	83.58	87.91	79.82	85.69	76.96
Ours	–	**90.68**	**83.93**	**89.83**	**84.87**	**89.76**	**84.05**	**86.28**	78.81

4 Experiments

4.1 Experiment Setup

Datasets. We validate our approach using four widely adopted benchmarks for class-incremental learning. For general image classification, we utilize CIFAR-100 [8], ImageNet-R [2], and ImageNet-100 [17]. To assess performance on fine-grained tasks, we include Split CUB-200 [20]. All datasets are divided evenly into sequences of 10 tasks and 20 tasks. This configuration facilitates a comprehensive analysis of the effectiveness of our method in both coarse- and fine-grained continual learning scenarios.

Evaluation Metrics. We report two metrics: (1) *Average Accuracy* (Avg), defined as the mean accuracy after each task, *i.e.*, $\frac{1}{T}\sum_{t=1}^{T} A_t$, where A_t represents the accuracy on all seen classes after task t; (2) *Last Accuracy* (Last), which corresponds to the final accuracy A_T upon completion of all tasks.

Implementation Details. We use CLIP ViT-B/16 as the backbone and inject LoRA modules with rank $r = 8$ into the query and value projections of each transformer layer. We train using AdamW optimizer with learning rate 1×10^{-4} and batch size 64. For multi-anchor distillation, we set the decay factor $\gamma = 0.9$ and retain a maximum of $M = 5$ anchors. The distillation loss weights are $\lambda_f = 1.0$ and $\lambda_a = 0.5$. For drift-aware early stopping, we use threshold $\tau = 0.1$. For the analytic classifier, we set $\lambda_1 = 0.1$ (ridge regularization), $\lambda_2 = 1.0$ (text

Table 2: Comparison with state-of-the-art methods under longer task sequences on CIFAR-100 and ImageNet-R. Best in **bold**, second best underlined.

		CIFAR-100				**ImageNet-R**			
		20 tasks		50 tasks		20 tasks		50 tasks	
Method	Venue	Avg	Last	Avg	Last	Avg	Last	Avg	Last
L2P [22]	CVPR'22	79.18	68.67	75.42	63.18	78.39	72.09	74.53	66.47
DualPrompt [21]	ECCV'22	79.74	69.91	76.05	64.23	79.12	71.69	75.18	65.92
CODA-Prompt [18]	CVPR'23	69.78	41.98	64.92	35.64	78.07	72.25	74.21	66.58
ACIL [34]	NeurIPS'22	78.56	70.23	74.18	65.42	76.82	69.54	72.36	64.18
RanPAC [13]	NeurIPS'23	81.24	73.56	77.85	69.12	79.45	72.38	75.62	67.85
AnaCP [14]	NeurIPS'24	83.42	76.18	80.25	72.34	81.12	74.26	77.48	69.52
PROOF [30]	TPAMI'25	85.12	76.13	81.47	70.85	80.25	73.42	76.18	68.25
RAPF [4]	ECCV'24	86.87	79.26	83.24	74.12	82.35	75.68	78.42	70.15
CLAP [6]	NeurIPS'24	86.08	77.35	82.51	72.48	81.92	74.85	77.65	69.38
BOFA [9]	AAAI'26	85.23	78.14	83.53	75.13	83.99	80.53	84.48	75.59
CLG-CBM [25]	CVPR'25	85.12	76.19	82.93	76.51	86.53	77.98	84.53	76.53
SECA [1]	AAAI'26	87.11	80.07	83.68	75.23	87.63	80.79	85.82	76.85
Ours	–	**88.61**	**82.95**	**85.47**	**78.32**	**88.97**	**83.60**	**86.28**	**78.94**

prior), and $\lambda_3 = 0.5$ (text subspace regularization). During inference, we set $\alpha = 1.0$ to scale the visual logit.

Competing Methods. We benchmark against three categories: (1) *Prompt-based methods* including L2P [22], DualPrompt [21], CODA-Prompt [18], SLCA [26], and Aper [28]; (2) *Analytic learning methods* including ACIL [34], RanPAC [13], and AnaCP [14]; and (3) *CLIP-based methods* including Continual-CLIP [19], PROOF [30], CLAP [6], RAPF [4], MOE4CL [24], MagMax [12], ENGINE [29], MG-CLIP [5], BOFA [9], CLG-CBM [25], and SECA [1]. All methods use ViT-B/16 from OpenAI [15]. For a fair comparison, all baseline results are reproduced under the same experimental setting using their official codebases with identical dataset splits, class orderings, and evaluation protocols.

5 Experimental Results

5.1 Comparison with State-of-the-Art Methods

Tables 1 and 2 present the comparative results under the 10-task, 20-task and 50-task settings, respectively. Key observations are summarized as follows.

Consistent improvements across benchmarks and settings. As shown in Tables 1–2, our method achieves the highest average accuracy on all four benchmarks under both 10-task and 20-task protocols. Rather than excelling on a particular dataset or metric, the improvements are broad-based, spanning coarse-grained (CIFAR-100, ImageNet-100), distribution-shifted (ImageNet-R), and fine-grained (CUB-200) scenarios alike. This generality suggests that the performance gains originate from structural advantages of the framework rather than dataset-specific tuning.

Robustness under longer task sequences. A notable trend emerges when comparing the 10-task, 20-task, and 50-task columns: while most base-

Table 3: Ablation studies of each component in our method on the three benchmarks.

				CIFAR-100		ImageNet-R		CUB-200	
Idx	M.A.D.	O.A.C.	C.M.R.	Last ↑	Avg ↑	Last ↑	Avg ↑	Last ↑	Avg ↑
1	–	–	–	75.15	66.68	79.12	72.00	67.00	54.80
2	✓	–	–	79.95	88.63	80.37	87.95	76.67	85.13
3	✓	✓	–	80.97	88.93	80.86	88.28	76.13	84.56
4	–	✓	✓	81.06	87.78	82.90	89.07	75.72	82.87
5	✓	✓	✓	**83.70**	**90.52**	**84.69**	**89.66**	**78.63**	**86.10**

lines degrade substantially as the number of tasks increases, our method exhibits markedly slower performance decay. The gap between our method and the strongest baseline in fact *widens* with longer sequences. This behavior is expected from our design: multi-anchor distillation provides long-range consistency constraints that prevent the compounding of small per-task drifts, and the analytic classifier is updated in closed form without revisiting old data, making it inherently immune to the accumulation of gradient-based overwriting errors. Frequent task transitions stress-test precisely these two failure modes, explaining why the relative advantage grows.

Encoder-level and classifier-level anti-forgetting are complementary. The tables reveal a clear hierarchy among method families. Prompt-based approaches (L2P, DualPrompt, CODA-Prompt) lag behind CLIP-based methods that explicitly address encoder drift (RAPF, CLAP, SECA), confirming that encoder-level protection is a prerequisite for strong continual learning performance. Yet even among the latter group, all methods still employ gradient-trained classifiers that are susceptible to boundary overwriting as new classes arrive. Our method closes this remaining gap by replacing the iterative classifier with a closed-form analytic solution, effectively decoupling the classifier from the forgetting dynamics of the encoder. The consistent margin over the best encoder-only methods across all settings validates the necessity of addressing both levels jointly.

Fine-grained recognition highlights the role of text subspace regularization. On CUB-200, where classes share high visual similarity, the overall ranking shifts noticeably compared to coarse-grained benchmarks. Prompt-based methods such as CODA-Prompt become more competitive in last-task accuracy, likely because prompt decomposition offers flexibility in assembling task-specific features at inference time. However, this comes at the cost of elevated forgetting across the full sequence, as reflected in the gap between average and last accuracy. Our method achieves the best average accuracy on this benchmark, suggesting that text subspace regularization helps the classifier capture fine-grained visual distinctions beyond what class-name semantics alone encode, while the analytic update rule preserves these distinctions as new classes are introduced.

5.2 Ablation Study

Component ablation. Table 3 validates each component of our method. Two key observations emerge. First, the encoder-side and classifier-side mechanisms

Table 4: Ablation on the text-guided analytic classifier's regularization terms on the three benchmarks.

			CIFAR-100		ImageNet-R		CUB-200	
Idx	$\mathcal{L}_{\text{prior}}$	$\mathcal{L}_{\text{sub}}$	Last ↑	Avg ↑	Last ↑	Avg ↑	Last ↑	Avg ↑
1	–	–	80.42	87.25	81.98	89.02	77.12	84.69
2	✓	–	82.15	88.91	82.98	89.15	77.93	85.59
3	–	✓	81.28	88.03	83.56	89.12	78.13	85.67
4	✓	✓	**83.70**	**90.52**	**84.69**	**89.66**	**78.63**	**86.10**

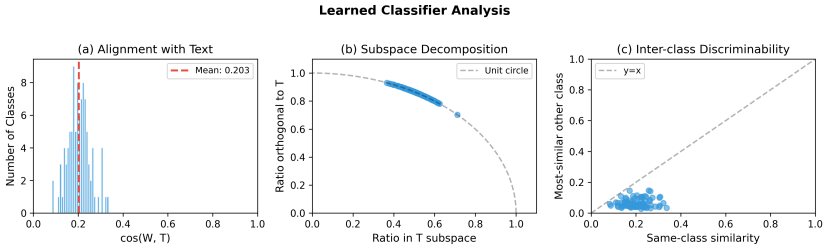
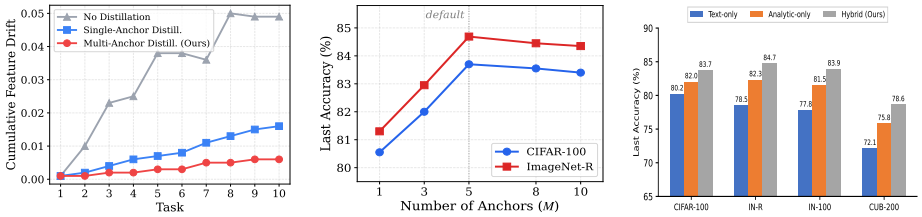


Fig. 2: Geometric analysis of learned classifier weights $\mathbf{W}$ on CIFAR-100. The classifier is largely orthogonal to text embeddings ($\cos(\mathbf{W}, \mathbf{T}) = 0.20$), with most energy in the orthogonal subspace (ratio ≈ 0.85).

address distinct bottlenecks. Multi-anchor distillation (M.A.D., Idx 2) primarily reduces cumulative feature drift but leaves the classifier vulnerable to overwriting, while the orthogonal analytic classifier with cross-modal regularization (O.A.C.+C.M.R., Idx 4) strengthens decision boundaries but cannot compensate for encoder drift. Neither alone matches the full method, confirming that the two forgetting sources are largely independent and must be addressed jointly. Second, cross-modal regularization plays a critical role beyond the base analytic solution. Comparing Idx 3 and Idx 5, where the only difference is whether C.M.R. (text prior and text subspace regularization) is included, the consistent improvement across all three benchmarks shows that injecting text semantics and encouraging orthogonal visual directions provide complementary information that the closed-form solution alone does not capture. The full method (Idx 5) achieves the best results across all benchmarks.

Text-guided regularization analysis. Table 4 dissects the two regularization terms that define the text-guided analytic classifier, where $\mathcal{L}_{\text{prior}}$ denotes the text prior ($\lambda_2 \|\mathbf{W} - \mathbf{T}\|_F^2$) and $\mathcal{L}_{\text{sub}}$ denotes text subspace regularization ($\lambda_3 \|\mathbf{T}\mathbf{W}^\top\|_F^2$). Two key observations emerge. First, each term provides independent gains over the base analytic classifier (Idx 1): the text prior (Idx 2) contributes more than text subspace regularization alone (Idx 3), indicating that a semantically grounded anchor is the stronger single component. Second, combining both (Idx 4) yields improvements that exceed either term in isolation across all three benchmarks, confirming their complementary roles: $\mathcal{L}_{\text{prior}}$ provides a semantic base, while $\mathcal{L}_{\text{sub}}$ steers the classifier toward orthogonal visual patterns that text semantics cannot capture.



(a) Cumulative feature drift. **(b)** Sensitivity to anchor count. **(c)** Classifier branch analysis.

Fig. 3: Analysis on CIFAR-100 (10 tasks). (a) Cumulative feature drift under different distillation strategies. (b) Sensitivity of anchor count M; performance saturates around M=5. (c) Last accuracy of each classifier branch across four benchmarks.

Cumulative drift analysis. As shown in Figure 3a, cumulative feature drift after 10 tasks is 0.050 without distillation, 0.016 with single-anchor distillation, and only 0.006 with our multi-anchor variant. The persistent growth under single-anchor distillation indicates that constraining only the preceding task is insufficient, as small per-task drifts compound over the learning sequence. By maintaining anchors across multiple historical states, multi-anchor distillation effectively suppresses this compounding effect.

Anchor count and classifier branch analysis. Figure 3b varies the anchor count M. Accuracy improves progressively from M=1 to M=5, confirming the benefit of multi-anchor constraints, while further increases offer marginal gains at higher storage cost. This saturation is consistent with our exponential decay weighting that naturally attenuates distant anchors. Figure 3c compares the per-task Last accuracy of each classifier branch. The text branch provides stable but lower accuracy, while the visual classifier offers stronger discriminative power that improves as more tasks are learned. Their hybrid combination consistently outperforms either branch alone, confirming that text embeddings and the learned classifier encode complementary cues.

6 Conclusion

We presented a unified exemplar-free framework for CLIP-based class-incremental learning that addresses forgetting at both the encoder and classifier levels. Multi-anchor distillation bounds cumulative feature drift by enforcing consistency with multiple historical encoder states, while the text-guided analytic classifier eliminates classifier-level forgetting by deriving weights in closed form from incrementally accumulated sufficient statistics. Text subspace regularization further encourages the classifier to capture complementary visual patterns that text semantics alone cannot provide. Experiments across four benchmarks under 10-, 20-, and 50-task settings demonstrate consistent improvements over prior methods, with the advantage becoming more pronounced as task sequences grow longer.

Limitations. The current framework stores multiple encoder checkpoints for multi-anchor distillation, incurring linear memory overhead in the number of

anchors. Additionally, the analytic classifier assumes a linear decision boundary, which may limit expressiveness on tasks requiring highly nonlinear separation. Future work could explore compressed anchor representations and nonlinear analytic formulations to address these constraints.

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